

**The Faculty of Engineering  
The Maersk Mc-Kinney Moller Institute  
Report/Assignment Cover Page**

|                      |                           |
|----------------------|---------------------------|
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|                             |                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------|
| Class and Year:             | Master of Science in Engineering Drone Technology and Autonomous Systems 2025–2027 |
| Subject Code and Name:      | T550021101 Kunstig intelligens værktøjer, (F26)                                    |
| Lecturer Name(s):           | Smith Khare and Vinay Chakravarthi Gogineni                                        |
| Title of Report/Assignment: | PAPER TITLE                                                                        |
| Submission Date:            | 04-06-2026                                                                         |

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**Contents**

|          |                                   |          |
|----------|-----------------------------------|----------|
| <b>1</b> | <b>Introduction</b>               | <b>1</b> |
| <b>2</b> | <b>Related Works</b>              | <b>1</b> |
| <b>3</b> | <b>Methodology</b>                | <b>1</b> |
| 3.1      | Model architecture . . . . .      | 1        |
| <b>4</b> | <b>Results</b>                    | <b>1</b> |
| <b>5</b> | <b>Discussion and Conclusions</b> | <b>1</b> |
| <b>6</b> | <b>Reflection</b>                 | <b>2</b> |

## **Abstract**

This paper presents the implementation and training of a convolutional neural network (CNN) in combination with a long short term memory (LSTM) model, for detection of active seizures in electroencephalogram (EEG) signals. The model operates on minimally preprocessed data, which enables easy and, more importantly, fast preprocessing, so that the model could potentially be used in a real-time context. The model hyper-parameters were tuned via a grid-search algorithm, leading to a highly performant model. The resulting model was compared to a simpler machine learning model, both on the same dataset, but also on a more complex dataset to show that the more advanced model is more generalisable.

*Please note that the title page, abstract, table of contents, and pages containing references are not included in the final report length. Ensure that the main body of the report does not exceed 7-8 pages.*

# 1 Introduction

This section should clearly describe your selected task. For example, if your task is image classification, begin by explaining what image classification is, why is its significance, and where it is commonly applied in real-world scenarios.

About  $\frac{1}{2}$  page.

# 2 Related Works

You should describe the state-of-the-art (SOTA) deep neural networks relevant to your task. After presenting these methods, clearly justify the motivation behind choosing your specific algorithm. For example, if you selected MobileNet for an image classification task, explain why MobileNet was preferred over other SOTA models, highlighting its advantages in the context of your application.

About 1 page.

# 3 Methodology

## 3.1 Model architecture

Describe the model architecture, dataset, pre-processing steps (if any), training strategy details (optimizer, loss, etc.), evaluation metrics, and implementation details. Please make sure that all the figures should be clear and must have a caption. Further, equations should be numbered correctly. Also, all the variables in the equations should be explained.

Example of equation with numbering:

$$E = mc^2 \tag{1}$$

Example of figure insertion:

Figure 1: Caption describing your figure.

About 2 pages.

# 4 Results

Summarize model performance, loss, accuracy (vs epoch) plots, confusion matrix, and any key visualizations. Make sure you compare your results with at least one of the SOTA algorithm. You are allowed to compare your results with your other group member results (provided you are implementing different algorithms/models). Please make sure that the plots should have x and y labels along with their units (if any).

About 2 pages.

# 5 Discussion and Conclusions

Discuss findings, implications, and limitations.

About  $\frac{1}{2}$  page.

## **6 Reflection**

Briefly summarize your experience in implementing this task and your observations/achievements, and future directions.

About  $\frac{1}{2}$  page.

**References**